Speed Automation System for Accident-Prone Zones Using IoT and Embedded Control

Abstract-Over-speeding in restricted and accident-prone zones such as school areas, hospitals, sharp curves, and construction sites is a major cause of road accidents. This paper presents an RFID-based automatic speed control system using a motor driver, designed to automatically regulate vehicle speed when entering predefined restricted zones. The proposed system utilizes RFID technology to identify zone-specific speed limits, a microprocessor to process control logic, and a motor driver to enforce speed regulation through PWM-based motor control. By integrating zone detection, closed-loop speed control, and hardware-level enforcement, the system minimizes driver dependency and enhances road safety. Experimental evaluation confirms reliable zone detection, smooth speed transition, and effective speed enforcement, making the system suitable for intelligent transportation and accident prevention applications.

Index Terms RFID, automatic speed control, motor driver, microprocessor, PWM, accident-prone zones.

I. INTRODUCTION

Road traffic accidents remain a critical public safety concern, particularly in developing countries where traffic rule enforcement is often inconsistent. Accident-prone zones such as school zones, hospital areas, sharp turns, and road construction sites require strict speed regulation to protect pedestrians and drivers. Conventional methods such as static signboards and manual policing depend heavily on driver compliance, which is often unreliable.

Advances in embedded systems and wireless identification technologies have enabled the development of intelligent speed control systems. RFID technology offers a low-cost and reliable solution for zone identification, while microprocessor-based control enables real-time decision-making. When combined with a motor driver, the system can automatically enforce speed limits at the hardware level.

This paper proposes an RFID-based automatic speed control system where RFID tags installed in restricted zones transmit speed limit information to a vehicle-mounted RFID reader. The microprocessor processes this information and controls motor speed through a motor driver using PWM signals. The main objectives of this work are:

1. To detect accident-prone zones using RFID technology
2. To automatically regulate motor speed using a motor driver
3. To reduce driver dependency and human error
4. To improve road safety through embedded automation.

II. LITERATURE REVIEW

Automatic speed control has been an active area of research in the field of intelligent transportation systems due to its potential to reduce road accidents and improve pedestrian safety. Researchers have explored various technologies to monitor vehicle speed and enforce speed limits in sensitive zones such as school areas, hospitals, residential streets, and pedestrian crossings.

Early studies on speed control systems mainly relied on manual enforcement and warning-based approaches. These systems typically used visual indicators, buzzers, or dashboard alerts to inform drivers when they exceeded speed limits. Although such methods increased driver awareness, their effectiveness was limited because they still depended on human reaction and decision-making. In many cases, drivers ignored warnings or reacted too late, resulting in accidents.

With the advancement of embedded systems, researchers began integrating microcontrollers with sensors and actuators to automate speed regulation. Several studies have demonstrated that microcontroller-based control systems are capable of adjusting vehicle speed in real time by processing environmental inputs and executing control logic efficiently. These systems significantly reduce human error and ensure consistent enforcement of speed limits.

RFID technology has been widely used in transportation-related applications such as electronic toll collection, vehicle identification, parking management, and access control. Due to its contactless operation, low power consumption, and ease of integration, RFID has attracted attention for location-based speed control applications. Researchers have shown that RFID-based systems can accurately detect predefined zones and trigger appropriate control actions without requiring complex computations.

Some studies have proposed GPS-based speed control systems where vehicle speed is regulated based on geographical location. While GPS provides wide-area coverage, it suffers from several limitations, including reduced accuracy in urban environments, signal loss, higher cost, and increased system complexity. Compared to GPS, RFID-based solutions offer precise zone detection within short ranges and are more suitable for specific locations such as school zones.

Other research works have explored sensor-based approaches using ultrasonic sensors, infrared sensors, or cameras to detect pedestrians and obstacles. These systems are effective for collision avoidance but may generate false detections due to environmental factors such as lighting conditions, weather, or moving objects. Moreover, many of these systems focus primarily on obstacle detection rather than enforcing speed limits based on location.

Recent literature emphasizes the importance of integrated systems that combine automatic zone detection with direct speed control mechanisms. Studies indicate that systems capable of automatically reducing speed without driver intervention are more reliable and effective in preventing accidents in high-risk areas. However, many existing solutions are either expensive, complex, or unsuitable for educational implementation.

Based on the reviewed literature, it is evident that there is a need for a simple, low-cost, and reliable automatic speed control system that can be easily implemented using basic microprocessor concepts. The proposed RFID-based speed control system addresses these limitations by providing accurate zone detection, automatic speed regulation, and real-time feedback using minimal hardware and computational resources.

III. METHODOLOGY

This section describes in detail the design approach, hardware–software integration, and operational logic of the proposed RFID-based automatic speed control system. The methodology focuses on implementing a simple yet effective control mechanism using basic microprocessor principles suitable for educational and practical applications.

A. Overall System Operation

The proposed system follows a zone-based speed control approach. The core idea is to automatically regulate vehicle speed based on the detection of predefined RFID tags that represent different traffic zones. Two RFID tags are used in the system: one representing a **school zone** and another representing a **normal zone**. Each tag has a unique identification number that allows the system to distinguish between restricted and unrestricted areas.

When the system is powered on, the Arduino UNO initializes all connected peripherals, including the RFID reader, motor driver, LCD display, and control switch. The system then continuously monitors the RFID reader for tag detection. This continuous scanning ensures real-time response when a vehicle enters or exits a school zone.

B. RFID-Based Zone Detection

The RFID reader module (RC522) operates using radio frequency communication and is interfaced with the Arduino UNO through the SPI protocol. When an RFID card or key is brought near the reader, the module reads the unique identification code of the tag and sends it to the microcontroller. The Arduino compares the received ID with stored reference values.

If the detected ID matches the school-zone tag, the system switches to restricted-speed mode. If the detected ID matches the normal-zone tag, the system switches back to normal-speed mode. This comparison process is fast and reliable, allowing immediate decision-making without noticeable delay.

C. Motor Control and Speed Regulation

A DC motor is used to simulate vehicle movement in the prototype. The motor is driven through a motor driver module (L298N or L293D), which acts as an interface between the low-power Arduino signals and the higher power required by the motor. This ensures electrical isolation and protects the microcontroller from high current loads.

Motor speed is controlled using Pulse Width Modulation (PWM). In normal-zone mode, the Arduino outputs a higher PWM duty cycle to the motor driver, allowing the motor to rotate at normal speed. In school-zone mode, the PWM duty cycle is reduced, thereby limiting the motor speed. This method allows smooth speed transition rather than sudden stopping, which is important for safety and mechanical stability.

IV. SYSTEM ARCHITECTURE

The system architecture of the proposed RFID-based automatic speed control system describes how different hardware modules are organized and how they interact to achieve automatic speed regulation in school zones. The architecture is designed to be simple, modular, and reliable, making it suitable for educational implementation and real-world extension.

The complete system is divided into three main functional units: the **Input Unit**, **Processing Unit**, and **Output Unit**. These units work together to detect the vehicle's location, process decision logic, and control speed accordingly.

A. Input Unit

The input unit is responsible for collecting real-time information required for system operation. It consists of an **RFID reader module** and a **manual ON/OFF switch**.

The RFID reader continuously scans for RFID cards or tags placed near the vehicle path. Each RFID tag represents a specific traffic zone. A school-zone RFID card indicates a restricted area, while a normal-zone RFID tag represents an unrestricted area. When a tag is detected, the RFID reader reads its unique identification number and sends this data to the microcontroller.

The ON/OFF switch provides manual control over the system. It allows the user to start or stop the motor operation at any time, ensuring safety and convenience during testing or emergency situations.

B. Processing Unit

The processing unit is the core of the system and is implemented using an Arduino UNO microcontroller. The Arduino receives input data from the RFID reader and the control switch and processes this information based on the programmed logic.

The microcontroller compares the detected RFID tag ID with predefined reference IDs stored in its memory. If the detected tag corresponds to a school zone, the Arduino switches the system to reduced-speed mode. If the detected tag corresponds to a normal zone, the Arduino restores the motor speed to its default value.

The Arduino also generates Pulse Width Modulation (PWM) signals to control the motor speed through the motor driver. This centralized processing ensures fast decision-making and consistent system behavior.

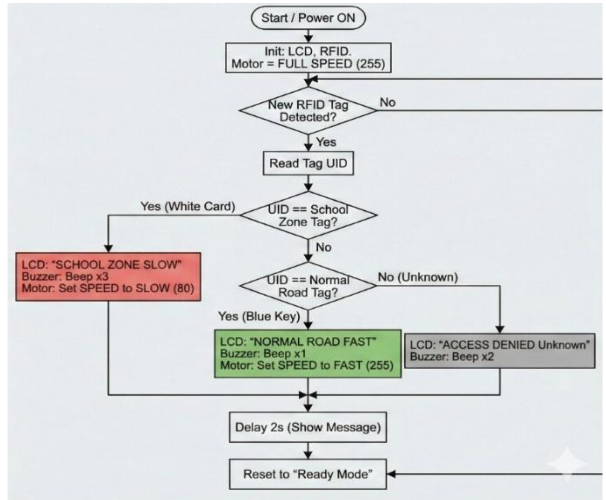

Fig: System Logic Flow Chart

C. Output Unit

The output unit executes the decisions made by the processing unit. It consists of a **motor driver module**, a **DC motor**, and an **LCD display**.

The motor driver acts as an interface between the low-power Arduino signals and the higher current required by the DC motor. Based on the PWM signals received from the Arduino, the motor driver adjusts the motor speed smoothly and safely. The DC motor represents the vehicle in the prototype model. Its speed changes automatically depending on the detected traffic zone.
The LCD display provides real-time visual feedback to the user. It displays system messages such as "School Zone – Speed Reduced" and "Normal Zone – Speed Restored," allowing easy monitoring of system status during operation.

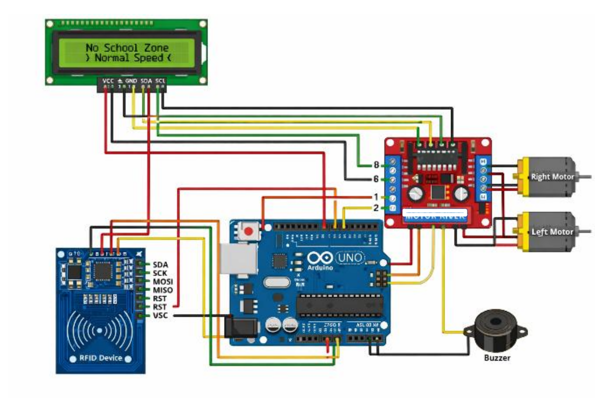

Fig: Circuit

D. Data and Control Flow

The data and control flow of the system follows a sequential and logical process. First, the RFID reader detects a nearby tag and sends the tag information to the Arduino. The Arduino processes this data, determines the current zone, and generates appropriate control signals. These signals are then sent to the motor driver to regulate motor speed, while the LCD is updated to display the current system state.

This structured flow ensures reliable operation, quick response time, and effective enforcement of speed limits in school zones.

V. RESULTS AND DISCUSSION

This section presents the experimental results obtained from the implementation of the RFID-based automatic speed control system and discusses the observed system performance under different operating conditions.

A. Results

The proposed system was tested using multiple RFID scans to verify its response accuracy, speed control behavior, and reliability. A DC motor was used to represent vehicle movement, and its speed variation was observed based on RFID detection.

When the normal-zone RFID tag was scanned, the system operated in normal mode. The DC motor rotated at its predefined maximum speed, and the LCD displayed the message "Normal Road – Speed Fast." This confirmed that the system correctly identified the unrestricted zone and allowed full-speed operation.

When the school-zone RFID card was scanned, the system immediately switched to restricted-speed mode. The Arduino reduced the PWM duty cycle supplied to the motor driver, resulting in a noticeable reduction in motor speed. At the same time, the LCD displayed "School Zone – Speed Slow," clearly indicating the change in operating condition.

The transition between normal speed and reduced speed was smooth, without abrupt stopping or instability. The ON/OFF switch successfully controlled motor operation at all times, allowing manual shutdown when required. The system responded consistently during repeated tests, showing reliable RFID detection and stable speed control.

B. Discussion

The experimental results demonstrate that the proposed system effectively implements automatic speed control based on location detection using RFID technology. By eliminating the need for driver intervention, the system reduces the possibility of human error, which is a major cause of accidents in school zones.

One of the key advantages observed is the **fast response time** of the system. The speed reduction occurred almost instantly after RFID detection, ensuring timely enforcement of speed limits. The use of PWM-based speed control allowed gradual and controlled speed changes, which is safer and more realistic than sudden braking.

The LCD display played an important role in improving system transparency by providing real-time feedback. This feature is especially useful during demonstrations, testing, and educational use, as it allows users to clearly understand the system's current state.

Although the prototype uses a DC motor instead of a real vehicle engine, the working principle remains the same. With appropriate actuators and integration, the system can be extended to real vehicles. However, the current system depends on RFID tag placement, which means proper infrastructure installation is required for real-world deployment.

Overall, the results confirm that the proposed RFID-based speed control system is **reliable, cost-effective, and suitable for school zone safety applications**, particularly as an educational and microprocessor-based project.

VI. REFERENCES

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